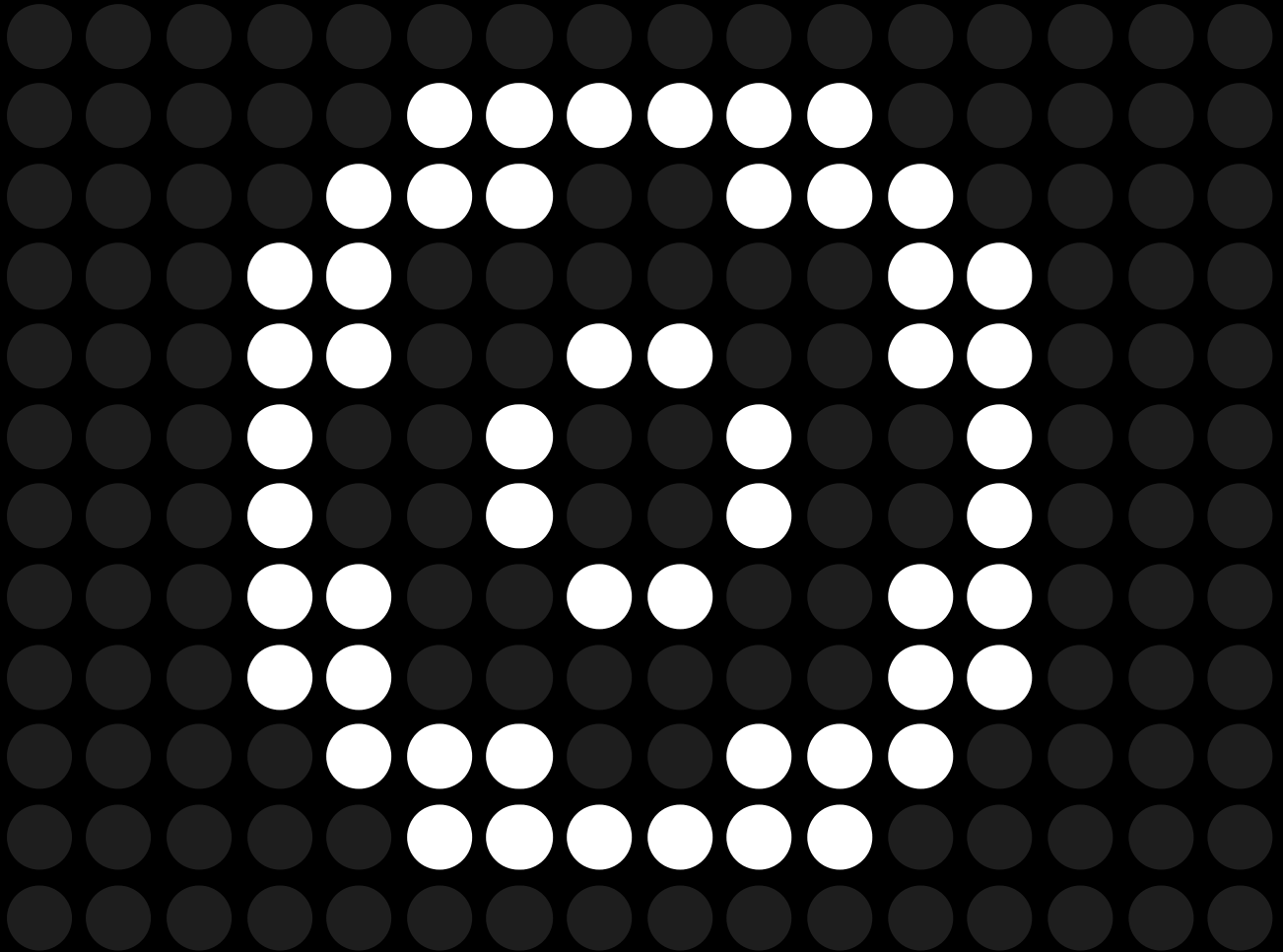




LUMEN

A 64×64 wall of light that runs on its own, day and night. You conduct it from your phone; the Raspberry Pi keeps the show going when the phone is put away.

FROM BOX TO GLOW

01	The system at a glance — who does what
02	The hardware — the full list, and the optional extras
03	The wiring — panel, controller, power
04	WLED — get the panel on your network
05	The Raspberry Pi — a headless system, ready to go
06	Install LUMEN — three commands
07	Use it — the app, its animations and modes
08	Voice control — optional, with an Amazon Echo
09	Update — from the Pi, or from the phone
10	Troubleshooting — if it gets stuck

ONE CONVENTION

Commands are typed in two windows. **Computer** means your computer's terminal. **Pi** means the Raspberry Pi's terminal, once you are connected to it (its prompt reads `you@lumen:~ $`). Each block says which one to use.

A NOTE ON LABELS

The app's controls are in French. Their names are kept here exactly as they appear on screen, with a translation alongside.

01 THREE PARTS

The Pi is the heart of the object. It computes every frame itself and sends it to the panel. The phone is only a remote: close it, and the panel keeps going.

The panel	64×64 LEDs. The glowing surface. It computes nothing, it displays.
The controller	A small board running WLED that physically drives the panel. It receives the images over the network.
The Raspberry Pi	Generates the animations, around the clock, and streams them to the controller. It also serves the app to your phone and remembers your settings and scenes.

AN IMAGE'S JOURNEY

The Pi paints a 64×64 frame → sends it over Wi-Fi to the controller (DDP protocol) → the controller lights the panel. Your phone only tells the Pi what to paint.

02 THE FULL LIST

Panel	HUB75 LED panel, 64×64.	Webcam	Optional. USB, for the camera animations (tested: Logitech C270).
Controller	Huidu HD-WF2 (ESP32-S3) running WLED.		
Raspberry Pi	3 A+ or better. 512 MB is enough.	Microphone	Optional. USB, for the sound modes.
Power	5 V / 8 A supply for panel and controller. The Pi has its own.		
microSD	For the Pi's system. 16 GB is plenty.	Amazon Echo	Optional. 2nd or 3rd gen, for voice control.

03 PANEL AND CONTROLLER

Everything is done with the power off. Plug in the supply only at the very end, once every connection is checked.

01

The data ribbon

Connect the HUB75 ribbon between the controller's X1 output and the panel's input (marked "IN", arrow pointing into the panel).

02

The panel's 5 V

Connect the supply's 5 V terminals to the panel's power terminals, respecting + and –.

03

The controller's 5 V

Power the controller from the same supply. Check the polarity once more.

04

Power on

Plug the supply in. The controller boots, the panel initialises.

POLARITY

Reversing + and – on the 5 V is the one real way to damage the hardware. When in doubt, check again before plugging in.

04 GET THE PANEL ONLINE

01

Join the controller

On first boot it creates a Wi-Fi network called WLED-AP. Connect your phone to it and the WLED page opens.

02

Put it on your network

In WLED's Wi-Fi settings, enter your home network's name and password. The controller joins the same network as the Pi.

03

Declare the matrix

In WLED's LED / 2D settings, set a **64 × 64** matrix. WLED's test effects should now fill the panel.

04

Note its name

Write down the controller's network name (`wled-xxxxxx.local`) or its IP address. You will give it to LUMEN in step 07. The name is better: it survives a change of network.

KEEP YOUR NETWORK TO YOURSELF

Your Wi-Fi name and password go into WLED and into Raspberry Pi Imager — nowhere else. Neither LUMEN nor this guide ever needs them.

05 A HEADLESS SYSTEM

A light system without a desktop, configured in advance: network, remote access, name. Everything is done from your computer with Raspberry Pi Imager.

01

Install Raspberry Pi Imager

From [raspberrypi.com](https://www.raspberrypi.com). Insert the microSD card into your computer.

02

Choose the model and the system

Your Raspberry Pi model, then **Raspberry Pi OS Lite (64-bit)**.

03

Fill in the settings

(the gear, "Edit settings") before writing:

Hostname	lumen — the Pi will answer at <code>lumen.local</code> .
SSH	Enabled, with password authentication.
Username	A name and a password of your choice. The install script assumes laurent : use it, or adapt the script (step 06).
Wi-Fi	Your home network, the same one the controller joined.
Time zone	Your own — the "day" mode follows the real hour.

04

Write the card

Wait for the verification, put the card in the Pi and power it on.

05

Connect

A minute or two later, from your computer:

TERMINAL

COMPUTER

```
% ssh laurent@lumen.local
```

FIRST CONNECTION

SSH asks you to confirm the machine's fingerprint: type **yes**, then your password. The prompt changes to `laurent@lumen:~ $` — you are on the Pi.

06 THREE COMMANDS

The code is public on GitHub. You fetch it straight onto the Pi, then a script turns it into a service that starts on its own at every boot.

FETCH AND INSTALL

PI

```
$ sudo apt-get install -y git
$ git clone https://github.com/lnyssen/lumen.git ~/lumen
$ cd ~/lumen && ./setup-service.sh
```

WHAT THE SCRIPT DOES

numpy	Installs it if needed (once, about a minute). The animation engine relies on it to stay fluid on a small Pi.
lumen	The main service: the animation engine and the server on port 8000 . Restarted automatically if it ever stops.
lumen-alexa	The voice-control service (step 08). Harmless if you have no Echo.
Update rights	Lets the app restart the service after an update sent from the phone (step 09).

CHECK THAT IT RUNS

PI

```
$ systemctl status lumen
```

You should read **active (running)**. From then on, the Pi runs the show by itself.

ANOTHER USERNAME?

`setup-service.sh` assumes the user **laurent** and the folder `/home/laurent/lumen`. If you chose another name, replace both in the script before running it.

07 FROM YOUR PHONE

01

Same network

Your phone must be on the same Wi-Fi as the Pi and the controller.

02

Open the app

In the browser, go to `http://lumen.local:8000` (or the Pi's IP address, port 8000).

03

Point it at the panel

Open **Adresse du panneau** (panel address) and enter the controller's name or IP from step 04. The top of the screen then reads **En ligne** (online).

04

Keep it on the home screen

Share → **Add to Home Screen**. LUMEN then opens full-screen, like an app.

THE SCREEN

A live preview of the panel, then the **Animation** and the **Palette**, and the sliders that shape the current animation (speed, scale, blend...). Along the bottom:

Pause	Freezes the current frame.
Veille	Sleep: switch off at a set time, or after 15 min, 30 min, 1 h.
Aléatoire	Random: LUMEN changes animation on its own.
Alimentation	Power: on / off.
Journée	Day: the mood follows the real hour, from dawn to night.
Scènes	Scenes: save the current look in a slot, recall it with a tap.
Lumière	Light: brightness, automatic adaptation to the room's light, Ambiance sonore (the object follows the room's mood: lively when it moves, calm in the silence), panel address and updates.

NEARLY FORTY, IN FAMILIES

Son • sound	Spectre, Pulse, Onde — they listen through the microphone.
Caméra • camera	Mirror, Silhouette, Trace — they watch through the webcam.
Créatif	Draw — paint the panel with your finger.
Géométrique	Plasma, Orb, Truchet, Moiré, Rings, Stripes, Glitch, Flow, Life, Checkers, Spiral, Star, Kaleido, Orbit.
Minimal	Sequence, Sweep, Bars, Static, Lines, Sine.
Math	Lissajous, Hilbert, reaction–diffusion, EKG.
Temps • time	Aujourd'hui (today), Weather, test card, Timecode, two clocks.
Texte	Your own words, scrolling.

Any animation can be worn in any of the **66 palettes**. The camera and sound families need the optional webcam and microphone; the others need nothing else.

08 WITH AN ECHO

Fully local: no cloud, no account to link. The Pi poses as a few smart plugs that the Echo finds on its own.

01

Discover

Once installed, say: *"Alexa, discover devices."* The LUMEN plugs appear.

02

Speak

Switch the panel on or off, or call a mode: day, random, music, plasma. The plugs carry French names (*Lumen, Journée, Aléatoire, Musique, Plasma*) — say them as written.

09 TWO WAYS

FROM THE PI

FETCH THE LATEST VERSION

PI

```
$ cd ~/lumen && git pull && ./setup-service.sh
```

FROM THE PHONE

Open **Lumière** → **Mettre à jour l'app depuis ce téléphone** (update from this phone), or go straight to `http://lumen.local:8000/update`. Choose the update's .zip file and confirm: the Pi installs it and restarts by itself. Reload the app five seconds later.

10 IF IT GETS STUCK

lumen.local not found	The Pi takes a minute or two to appear after power-on. Check it is on the right Wi-Fi; as a last resort, use its IP address.
Blank page	On the Pi, <code>systemctl status lumen</code> should read active (running). If not, run <code>./setup-service.sh</code> again.
Panel stays black	Check the 5 V and the ribbon on X1. In WLED, a test effect should light the panel on its own.
The app says offline	The panel address is wrong, or the controller changed IP: enter its <code>wled-xxxxxx.local</code> name instead.
Old version on the phone	Open the app once in a private tab, or remove and re-add the home-screen icon.
No camera / no sound	Check the USB webcam or microphone is plugged into the Pi. They switch on only when an animation needs them.

THE GOLDEN RULE

Go link by link: panel (does it light?), controller (does WLED answer?), network (same Wi-Fi?), Pi (is the service running?), app (right panel address?). Find the link, and you have found the fault.